

Module 3: Statistics

Descriptive Statistics, Frequentist Statistics and Mathematical Foundations

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Module 3: Overview

Statistics

- Descriptive statistics
- Sampling
- Estimation
- MSE
- Confidence intervals
- Parametric and non-parametric estimation

Mathematical Foundations

- Algebraic structures
- Vector spaces
- Vectors
- Dot product
- Orthogonality
- Hyperplanes

1. Introduction to Statistics

What is Statistics?

Definition

Statistics is the science of collecting, organizing, summarizing, analyzing and interpreting data.

Two major branches

- **Descriptive statistics** – summarizes observed data.
- **Inferential statistics** – draws conclusions about a population from a sample.

Data → Analysis → Inference → Decision

Data, Information and Knowledge



- **Data:** raw facts or observations.
- **Information:** processed data with meaning.
- **Knowledge:** understanding obtained from information and patterns.

Types of Data

Qualitative

- Categorical
- Nominal
- Ordinal

Examples:

Color, Grade, Department

Quantitative

- Numerical
- Discrete
- Continuous

Examples:

Age, Height, Number of Students

2. Descriptive Statistics

Descriptive Statistics

Descriptive statistics summarizes the important characteristics of observed data.

- Measures of central tendency
- Measures of dispersion
- Graphical summaries
- Distribution shape
- Outlier detection

Center + Spread + Shape

Measures of Central Tendency

Mean	Median	Mode
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- **Mean:** arithmetic average.
- **Median:** middle value after sorting.
- **Mode:** most frequently occurring value.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Sample Mean

For observations

$$x_1, x_2, \dots, x_n,$$

the sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Example

For 2, 4, 6, 8, 10:

$$\bar{x} = \frac{2 + 4 + 6 + 8 + 10}{5} = 6$$

Median and Mode

Median

- Sort the observations.
- Odd number of observations: middle value.
- Even number: average of two middle values.

Mode

- Most frequently occurring observation.
- More than one mode may exist.

2, 3, 3, 4, 7, 8, 10

Median = 4, Mode = 3

Measures of Dispersion

Dispersion measures how widely observations are spread.

Range	Variance	Standard Deviation	IQR
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$$\text{Range} = \max(x_i) - \min(x_i)$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s = \sqrt{s^2}$$

Sample Variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- Measures spread around the sample mean.
- Squaring prevents positive and negative deviations from cancelling.
- $n - 1$ is used for the usual unbiased estimator of population variance.

Population variance:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Variance and Standard Deviation

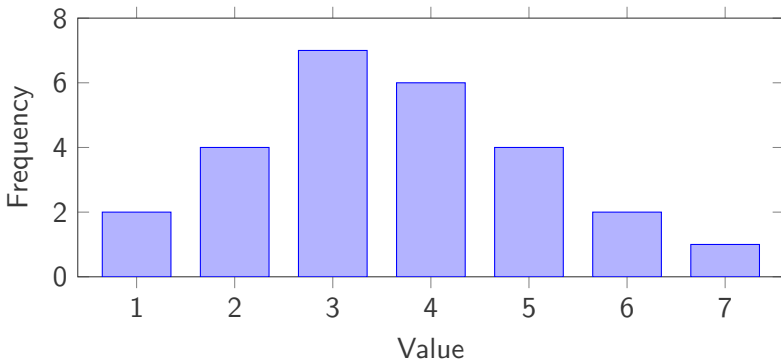
$$\text{Var}(X) = \mathbb{E}[X^2] - [\mathbb{E}(X)]^2$$

- Variance is measured in squared units.
- Standard deviation is measured in the original units.
- Higher variance indicates greater spread.

$$s = \sqrt{s^2}$$

Histogram

A histogram displays the frequency or density of numerical observations across intervals called **bins**.



Reading a Histogram

A histogram can reveal:

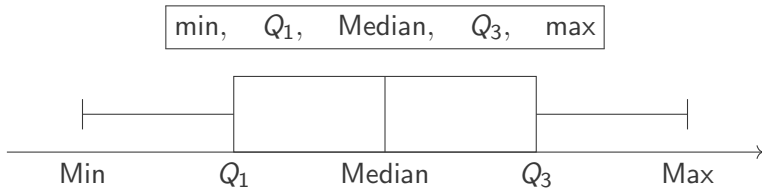
- Center
- Spread
- Skewness
- Number of modes
- Possible unusual observations

Important

The appearance of a histogram depends on the choice of bin width.

Boxplot

A boxplot summarizes data using:



$$IQR = Q_3 - Q_1$$

Outliers

An **outlier** is an observation unusually far from the majority of observations.
Using the IQR rule:

$$\text{Lower Fence} = Q_1 - 1.5IQR$$

$$\text{Upper Fence} = Q_3 + 1.5IQR$$

Observations outside these limits are commonly flagged as outliers.

Effect of Outliers

Outliers can strongly influence:

- Mean
- Variance
- Standard deviation
- Statistical models

More robust measures include:

Median and IQR

Key idea

A single extreme observation can significantly change the mean, while the median may remain relatively stable.

Kurtosis

Kurtosis describes the **tail heaviness** of a distribution.

$$\beta_2 = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}$$

- **Mesokurtic:** normal-like reference.
- **Leptokurtic:** heavier tails.
- **Platykurtic:** lighter tails.

Excess kurtosis:

$$\gamma_2 = \beta_2 - 3$$

Order Statistics

Given:

$$X_1, X_2, \dots, X_n$$

sort the observations:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

where $X_{(i)}$ is the i th order statistic.

- $X_{(1)}$ = minimum
- $X_{(n)}$ = maximum
- Middle order statistics determine the median.

Applications of Order Statistics

Order statistics are used for:

- Ranking
- Median calculation
- Quartiles
- Percentiles
- Boxplots
- Extreme-value analysis

Example:

7, 2, 9, 4, 1

1, 2, 4, 7, 9

Therefore:

$X_{(1)} = 1$ $X_{(2)} = 2$ $X_{(3)} = 4$ $X_{(4)} = 7$ $X_{(5)} = 9$

3. Covariance

Sample Covariance

For paired observations (x_i, y_i) :

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Interpretation:

- $s_{xy} > 0$: variables tend to increase together.
- $s_{xy} < 0$: one tends to increase as the other decreases.
- $s_{xy} \approx 0$: little linear co-movement.

Covariance and Variance

Variance is a special case of covariance:

$$\text{Cov}(X, X) = \text{Var}(X)$$

	Variance	Covariance
Variables	One	Two
Measures	Spread	Joint variation
Sign	Non-negative	+, - or 0

Sample Covariance Matrix

For

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix},$$

the covariance matrix is

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- Diagonal = variances
- Off-diagonal = covariances
- Symmetric matrix

Example: Covariance Matrix

Suppose:

$$\text{Var}(X_1) = 4, \quad \text{Var}(X_2) = 9, \quad \text{Cov}(X_1, X_2) = 3.$$

Then:

$$S = \begin{bmatrix} 4 & 3 \\ 3 & 9 \end{bmatrix}$$

The positive covariance indicates positive linear co-movement.

4. Algebraic Structures

Algebraic Structures

An algebraic structure consists of:

Set + Operation(s) + Axioms

A binary operation:

$$* : S \times S \rightarrow S$$

Important properties:

- Closure
- Associativity
- Identity
- Inverse
- Commutativity

Semigroup and Monoid

Semigroup

A set with a closed and associative binary operation:

$$(a * b) * c = a * (b * c)$$

Monoid

A semigroup with an identity element e :

$$e * a = a * e = a$$

Monoid = Semigroup + Identity

Group

A group $(G, *)$ satisfies:

1. Closure
2. Associativity
3. Identity
4. Inverse

For every $a \in G$:

$$\exists a^{-1} \in G$$

such that

$$aa^{-1} = a^{-1}a = e.$$

Example:

$$(\mathbb{Z}, +)$$

Abelian Group

An Abelian group is a group that is also commutative.

$$a * b = b * a$$

for all $a, b \in G$.

Example:

$$(\mathbb{Z}, +)$$

because:

$$a + b = b + a.$$

Key point

Every Abelian group is a group, but every group need not be Abelian.

Ring

A ring has two operations:

$$+, \quad \times$$

Typically:

$$(R, +)$$

is an Abelian group, while multiplication is associative and distributive over addition.

$$a(b + c) = ab + ac$$

$$(a + b)c = ac + bc$$

Example:

$$\boxed{\mathbb{Z}}$$

Field

A field is a commutative ring in which every non-zero element has a multiplicative inverse.

For $a \neq 0$:

$$\exists a^{-1}$$

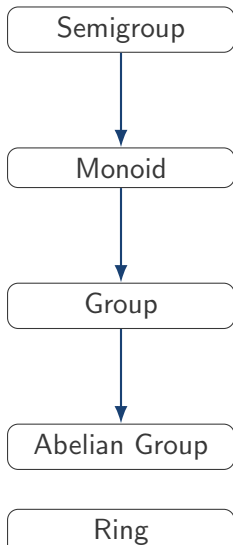
such that

$$aa^{-1} = 1.$$

Examples:

$$\boxed{\mathbb{Q}, \quad \mathbb{R}, \quad \mathbb{C}}$$

Algebraic Structure Hierarchy



5. Vector Spaces and Vectors

Mathematical Spaces

A mathematical space is a set of objects together with structure that allows relationships between them to be studied.

Examples:

- Metric spaces – distance
- Topological spaces – continuity and neighborhoods
- Vector spaces – vector operations
- Probability spaces – outcomes and probabilities

Vector Space

A vector space V over a field F supports:

- Vector addition
- Scalar multiplication

For $\mathbf{u}, \mathbf{v} \in V$ and $a, b \in F$:

$$a\mathbf{u} + b\mathbf{v} \in V$$

Important properties include:

- Additive identity $\mathbf{0}$
- Additive inverse $-\mathbf{v}$
- Associativity and commutativity of addition
- Distributivity of scalar multiplication

What is a Vector?

A vector has:

Magnitude + Direction

A vector in $\mathbb{R}^n$ can be written as:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Its Euclidean magnitude is:

$$\|\mathbf{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}$$

Scalar Multiplication

For a scalar c :

$$\mathbf{v}' = c\mathbf{v}$$

Magnitude:

$$\|c\mathbf{v}\| = |c| \|\mathbf{v}\|$$

- $|c| > 1$: magnitude increases.
- $0 < |c| < 1$: magnitude decreases.
- $c = 0$: zero vector.
- $c < 0$: direction reverses.

Changing the Magnitude of a Vector

Let

$$\mathbf{v} = (2, 1).$$

Then:

$$2\mathbf{v} = (4, 2)$$

has twice the magnitude.

$$\frac{1}{2}\mathbf{v} = \left(1, \frac{1}{2}\right)$$

has half the magnitude.

$$-\mathbf{v} = (-2, -1)$$

has the same magnitude but opposite direction.

Vector Addition

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix},$$

$$\boxed{\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}}$$

Vector addition is component-wise.

Impact of Vector Addition

The resultant depends on the angle between the vectors.

Same direction

$$\|\mathbf{u} + \mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$$

Opposite directions

Vectors can partially or completely cancel.

Perpendicular vectors

$$\|\mathbf{u} + \mathbf{v}\| = \sqrt{\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2}$$

Dot Product

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i$$

Geometrically:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

where θ is the angle between the vectors.

Orthogonal Vectors

Two non-zero vectors are orthogonal if:

$$\theta = 90^\circ.$$

Using:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta,$$

we get:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos 90^\circ$$

Therefore:

$$\boxed{\mathbf{u} \cdot \mathbf{v} = 0}$$

Proof of Orthogonality

Suppose:

$$\mathbf{u} \perp \mathbf{v}.$$

Then:

$$\theta = 90^\circ.$$

By the dot-product formula:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

Substitute $\theta = 90^\circ$:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| (0)$$

Hence: $\boxed{\mathbf{u} \cdot \mathbf{v} = 0}$

Two Aligned Unit Vectors

Let $\mathbf{u}$ and $\mathbf{v}$ be unit vectors:

$$\|\mathbf{u}\| = \|\mathbf{v}\| = 1.$$

If they are aligned:

$$\theta = 0^\circ.$$

Therefore:

$$\mathbf{u} \cdot \mathbf{v} = (1)(1) \cos 0^\circ$$

$$\boxed{\mathbf{u} \cdot \mathbf{v} = 1}$$

Hyperplane

A hyperplane in $\mathbb{R}^n$ can be represented by:

$$\mathbf{w}^T \mathbf{x} + b = 0$$

where:

- $\mathbf{w}$ = normal vector
- $\mathbf{x}$ = point
- b = bias/intercept

Examples:

$\mathbb{R}^2$: line

$\mathbb{R}^3$: plane

$\mathbb{R}^n$: $(n - 1)$ -dimensional hyperplane

Changing the Orientation of a Hyperplane

The vector $\mathbf{w}$ is normal to the hyperplane:

$$\mathbf{w}^T \mathbf{x} + b = 0.$$

Changing the direction of $\mathbf{w}$ changes the orientation of the normal and therefore the orientation of the hyperplane.

For a linear classifier:

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$$

the hyperplane:

$$f(\mathbf{x}) = 0$$

acts as the decision boundary.

Vectors and Hyperplanes in Machine Learning

A feature vector:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

is classified using:

$$\mathbf{w}^T \mathbf{x} + b.$$

- $\mathbf{x}$ – feature vector
- $\mathbf{w}$ – weight vector
- b – bias
- $\mathbf{w}$ – normal to decision boundary

Linear classifier $\longleftrightarrow$ Hyperplane

6. Frequentist Statistics

Frequentist Statistics

Frequentist statistics interprets probability through the behavior of repeated random sampling.

Population $\rightarrow$ Sample $\rightarrow$ Statistic $\rightarrow$ Inference

Parameter

$$\mu, \sigma^2$$

Fixed but generally unknown.

Statistic

$$\bar{X}, S^2$$

Calculated from a sample.

Sampling

Population

Complete collection of units or observations of interest.

Sample

Subset selected from the population.

Common approaches:

- Simple random sampling
- Stratified sampling
- Systematic sampling
- Cluster sampling

Sampling Distribution

A sampling distribution is the probability distribution of a statistic over repeated samples.

For the sample mean:

$$\mathbb{E}[\bar{X}] = \mu$$

and, under independent sampling,

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

As n increases, the sampling variability of $\bar{X}$ decreases.

Point Estimation

An **estimator** is a rule for estimating an unknown population parameter.

Example:

$$\hat{\mu} = \bar{X}$$

estimates μ .

Estimator vs Estimate

- Estimator – rule/random quantity.
- Estimate – numerical value obtained after observing data.

Mean Square Error

For an estimator $\hat{\theta}$:

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]$$

MSE measures expected squared estimation error.

It incorporates:

$$\text{Variance} + \text{Squared Bias}$$

Bias–Variance Decomposition

Bias:

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

MSE decomposition:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

Thus, a good estimator should balance bias and variability.

Consistency and Confidence Intervals

Consistency

An estimator is consistent if it approaches the true parameter as sample size increases:

$$\hat{\theta}_n \xrightarrow{P} \theta$$

Confidence interval

General form:

$$\text{Estimate} \pm \text{Critical value} \times \text{Standard Error}$$

Parametric and Non-Parametric Estimation

Feature	Parametric	Non-parametric
Assumptions	Stronger	Fewer
Parameters	Fixed/finite	Flexible
Flexibility	Lower	Higher
Example	Normal model	KDE

Parametric models are efficient when assumptions are appropriate.

Non-parametric methods are more flexible but may require more data.

Module 3 – Formula Summary

Mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

Variance: $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

IQR: $IQR = Q_3 - Q_1$

Covariance: $s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$

Dot product: $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$

Orthogonality: $\mathbf{u} \cdot \mathbf{v} = 0$

Vector scaling: $\|c\mathbf{v}\| = |c| \|\mathbf{v}\|$